

1. Consider a forward SDE

$$dx_t = f(x_t, t) dt + g(x_t, t) dW_t,$$

show that the corresponding probability flow ODE is written as

$$dx_t = \left[f(x_t, t) - \frac{1}{2} \frac{\partial}{\partial x} g^2(x_t, t) - \frac{g^2(x_t, t)}{2} \frac{\partial}{\partial x} \log p(x_t, t) \right] dt.$$

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Fokker–Planck equation: $\partial_t p(x, t) = -\partial_x (f(x, t)p(x, t)) + \frac{1}{2} \partial_x^2 (g^2(x, t)p(x, t))$

$$dx_t = v(x_t, t) dt$$

$$\partial_t p(x, t) = -\partial_x (v(x, t)p(x, t))$$

$$\Rightarrow \partial_x (v(x, t)p(x, t)) = \partial_x (f(x, t)p(x, t)) - \frac{1}{2} \partial_x^2 (g^2(x, t)p(x, t))$$

Integrating both sides: $v(x, t)p(x, t) = f(x, t)p(x, t) - \frac{1}{2} \partial_x (g^2(x, t)p(x, t))$

$$\Rightarrow v(x, t) = f(x, t) - \frac{1}{2p(x, t)} \partial_x (g^2(x, t)p(x, t))$$

$$= f(x, t) - \frac{1}{2p(x, t)} (p(x, t) \partial_x g^2(x, t) + g^2(x, t) \partial_x p(x, t))$$

$$\Rightarrow v(x, t) = f(x, t) - \frac{1}{2} \partial_x g^2(x, t) - \frac{1}{2} g^2(x, t) \partial_x \log p(x, t), \quad \frac{1}{p} \partial_x p = \partial_x \log p(x, t)$$

$$\therefore dx_t = [f(x_t, t) - \frac{1}{2} \partial_x g^2(x_t, t) - \frac{1}{2} g^2(x_t, t) \partial_x \log p(x_t, t)] dt$$

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In 20 years, AI might develop the ability to create high-fidelity virtual clinical trials, which simulate the biological, genetic, and behavioral complexity of human populations.

The system could generate digital twins of patients. It could test thousands of possible drugs or treatments virtually before any human testing. It could also predict side effects, drug interactions, and long-term health outcomes with near-human accuracy.

This would likely rely on methods such as supervised learning and reinforcement learning. Supervised learning can let it handle the connection between pairs of drug and the response. And Reinforcement learning enables it to learn which treatment strategies yield the best outcomes.

The data sources of this project are genomic databases, medical records, prior clinical trial data, biochemical simulations, and physiological models. The target signals are Survival rates, symptom reduction, and side-effect frequency. The system would continuously interact with a simulated biological environment, and test hypotheses, adjust treatment parameters, and observe how virtual patients respond over simulated time.

The first step of this is either an AI accurately predict drug efficacy and side effects using virtual patients in a small, simulated disease model. It'll need to focus on limited set of data to build up the digital pairs and compare to historical data to test if the model is successful.